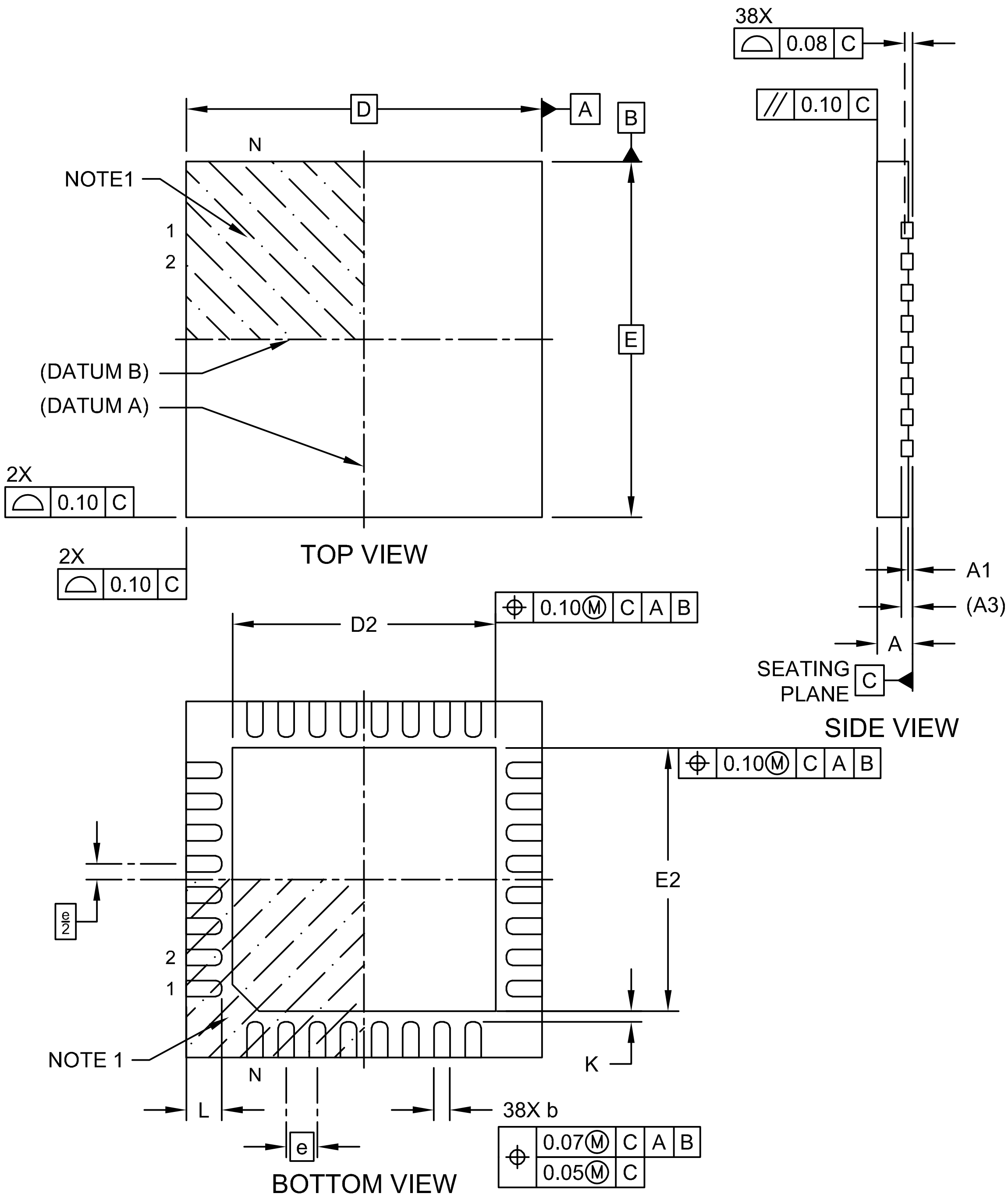
38-Lead Extremely-Thin Quad Flatpack, No Lead Package (TUB) - 4x4x0.4 mm Body [XQFN]; With 2.96x2.96mm Exposed Pad; Atmel Legacy GPC ZBQ

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



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38-Lead Extremely-Thin Quad Flatpack, No Lead Package (TUB) - 4x4x0.4 mm Body [XQFN]; With 2.96x2.96mm Exposed Pad; Atmel Legacy GPC ZBQ

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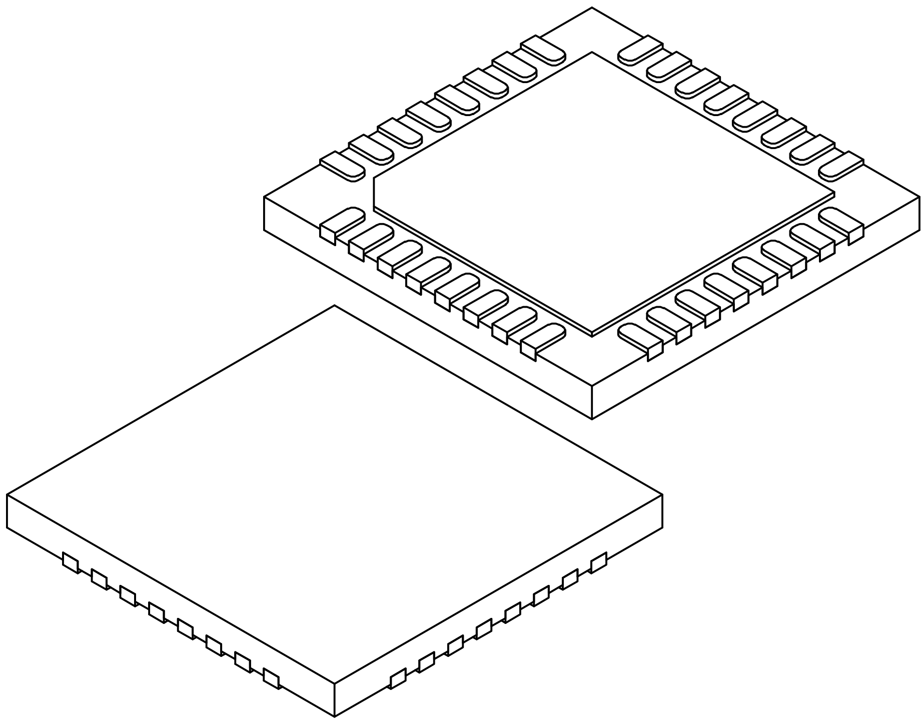


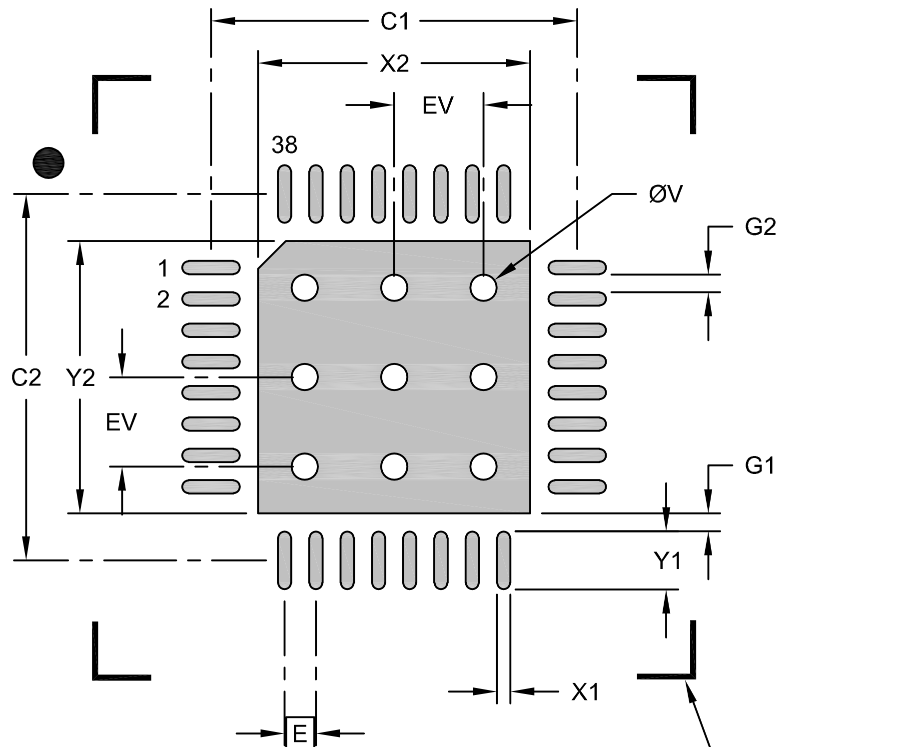
	Units	MILLIMETERS		
	Dimension Limits	MIN	NOM	MAX
Number of Terminals	N		38	
Pitch	e		0.350 BSC	
Overall Height	A	0.310	0.360	0.400
Standoff	A1	0.000	-	0.050
Terminal Thickness	A3		0.127 REF	
Overall Length	D		4.00 BSC	
Exposed Pad Length	D2	2.86	2.96	3.06
Overall Width	E		4.00 BSC	
Exposed Pad Width	E2	2.86	2.96	3.06
Terminal Width	b	0.130	0.180	0.230
Terminal Length	L	0.300	0.400	0.500
Terminal-to-Exposed-Pad	K	0.120	-	-

Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Package is saw singulated
3. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
   REF: Reference Dimension, usually without tolerance, for information purposes only.

38-Lead Extremely-Thin Quad Flatpack, No Lead Package (TUB) - 4x4x0.4 mm Body [XQFN]; With 2.96x2.96mm Exposed Pad; Atmel Legacy GPC ZBQ

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



RECOMMENDED LAND PATTERN

	Units	MILLIMETERS		
	Dimension Limits	MIN	NOM	MAX
Contact Pitch	E		0.350 BSC	
Center Pad Width	X2			3.05
Center Pad Length	Y2			3.05
Contact Pad Spacing	C1		4.10	
Contact Pad Spacing	C2		4.10	
Contact Pad Width (X38)	X1			0.15
Contact Pad Length (X38)	Y1			0.65
Contact Pad to Center Pad (X38)	G1	0.20		
Contact Pad to Contact Pad (X34)	G2	0.20		
Thermal Via Diameter	V		0.30	
Thermal Via Pitch	EV		1.00	

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
2. For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

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